

WRO Junior Future Innovators

HOME FARMER



*Written By
The Rooftop Farmers*



India



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Picture 1 - Yogeshwar & Vihaan

ABOUT US

The Rooftop Farmers, a duo of school students, are on a mission to revolutionize sustainable development through robotics. With a keen interest in environmental conservation, they're making innovative solutions that promise a greener future. These two young minds are tirelessly developing a sustainable robotics project, aiming to safeguard the environment and promote eco-friendly practices. Their dedication, creativity, and passion for sustainability make them a force to be reckoned with, embodying the spirit of youth-driven change in the quest for a more sustainable world.

VIHAAN

Loves Robotics, loves to create new things, and is also passionate about playing video games. Passion in coding makes him a code freak. He has made an app with his coding skills and is planning to develop even more. Other than coding, he likes to do light research and read books, along with singing.

YOGESHWAR

Interested in railroading, making show theories, aviation and flight simulations and also making flight simulation hardware. Is also interested in Lego models in Real life and also in BrickLink studios. Sometimes plays the piano, tennis and pickleball.



THE PROBLEMS

A Brief Description About the Current Scenario

The current scene regarding land pollution and soil infertility paints a concerning picture of environmental degradation. As the population grew, farmers started using chemical fertilizers to meet demands. This rampant use of chemical fertilizers has led to soil degradation, reduced fertility, and contamination of groundwater. These practices not only harm the soil but also affect the quality of our food supply. Excessive use of fertilizers contributes to nutrient imbalances in crops, leading to health issues like diseases of the kidneys and liver, and certain cancers in consumers. These adverse effects not only endanger public health but also hinder sustainable development by compromising the long-term viability of agricultural systems and ecosystems, emphasizing the urgent need for eco-friendly farming practices.

The Global Warming Effect

Global warming makes land pollution and soil problems worse. It changes rain and causes droughts or floods, harming soil. "High temperatures accelerate soil degradation, making food production more difficult and affecting health and sustainable development. We need to act fast to fix this.

The Urban Challenge

Urban food production is challenging due to limited space and busy lifestyles, making traditional gardening difficult." The problem matters as cities face a growing demand for sustainable, self-reliant food production.

THE SOLUTION



Problem Statement:

Our project, Home Farmer, aims to address the pressing issues of produce being grown with chemical fertilizers, and the growing need for sustainable urban farming. These challenges are critical due to their impact on food safety, environmental health, and sustainable development. Though “Very few people practice home farming because growing organic food at home is challenging and time-consuming.” Home farming is uncommon as people find it difficult to produce truly organic food without using chemical fertilizers.”



Why This Problem

We chose this problem because conventional farming practices often contribute to soil degradation and pollution through the heavy use of chemical fertilizers, pesticides, and intensive land cultivation. Urbanization further exacerbates these issues by reducing available arable land and increasing pollution levels.



Robotic Solution

Home Farmer is the robotic solution that we designed to revolutionize farming practices. It is capable of planting seeds, autonomously providing daily water requirements, and monitoring the environment in which the plants are growing. By incorporating sensors and sustainable farming techniques, Home Farmer optimizes resource usage and reduces reliance on harmful chemicals. The solution also reduces the human efforts for farming and makes it easily available at home.

THE SOLUTION



Value of the Robotic Solution

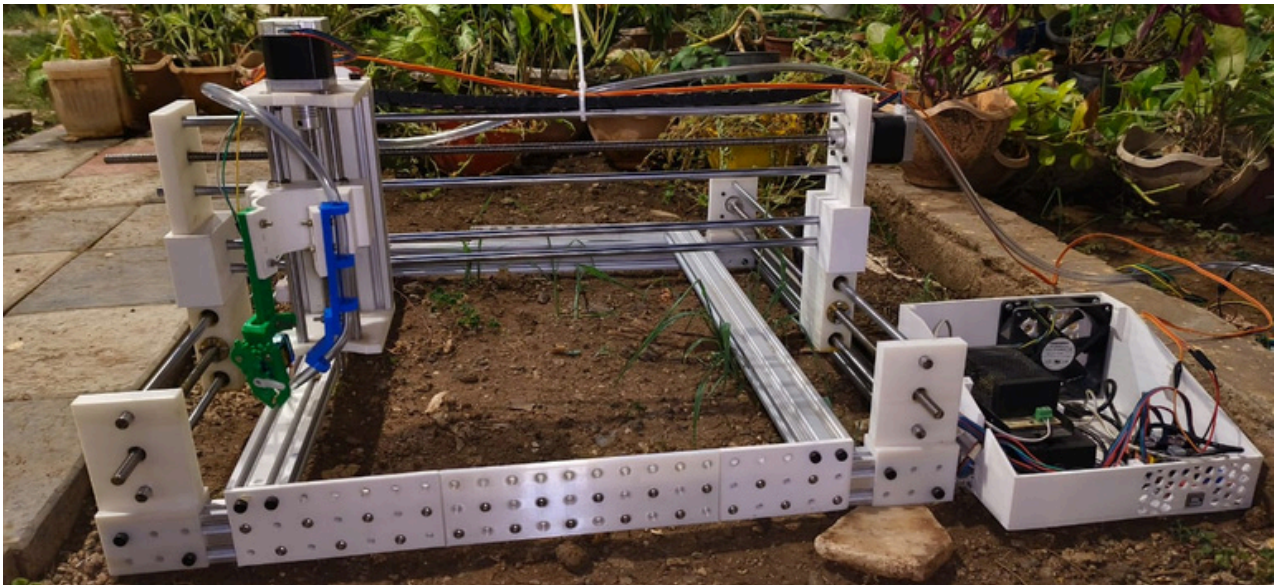
The implementation of Home Farmer in real-life scenarios offers immense value. It conserves water through precise irrigation, eliminates the need for chemical fertilizers, and promotes healthy plant growth. This not only improves crop yields but also enhances the quality of the plants. Furthermore, Home Farmer enables efficient urban farming, making it possible to produce fresh, healthy food. By dedicating minimal time to farming, individuals can enjoy freshly grown plants without the use of fertilizers, all conveniently at their doorstep. This is achievable without disrupting their daily schedules, work commitments, or leisure time, thanks to our home farmer



Importance of the Project

Home Farmer is crucial in addressing global challenges related to food production, environmental sustainability, and urban development. By promoting sustainable farming practices, reducing pollution, and conserving natural resources, our project contributes to building resilient agricultural systems and ensures a healthier future for both people and the planet.





Picture 1.2 - Home Farmer V1

HOME FARMER

Origins of the Idea: The concept of Home Farmer emerged from our deep concern about the environmental impact of traditional farming methods and the challenges posed by urbanization on agricultural practices. We recognized the need for a sustainable solution that could address soil degradation, land pollution, and the growing demand for urban farming.

Although established farming techniques exist, it's surprising that home farming isn't more popular. This is especially concerning given the harmful effects of pesticides on daily consumed crops. To tackle this issue, we came up with a simple and cost-effective solution – introducing Home Farmer.

OUR MISSION

To empower individuals with the tools and knowledge to cultivate their own food at home, we harness accessible and innovative technology. Our aim is to foster sustainability and self-sufficiency, guiding individuals towards healthier lifestyles and environmental stewardship.

OVERVIEW

Our project revolves around the development of a Home Farming Robot, a sophisticated yet user-friendly system designed to enable individuals to grow their own food at home autonomously. Leveraging cutting-edge technology such as robotics, automation, and sensor integration, the Home Farming Robot streamlines the process of home gardening, making it accessible to people of all skill levels.

OUR JOURNEY



How we got the idea for our solution

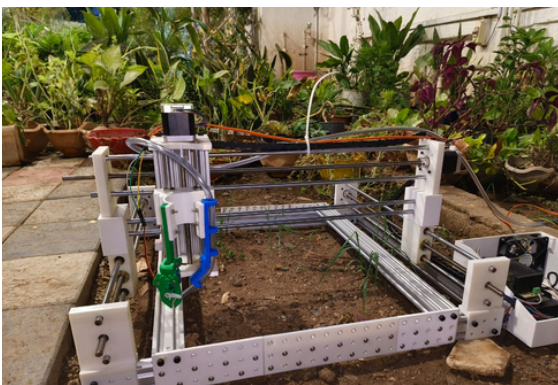


When we found a pricey commercial farming robot meant for home, we liked its features but couldn't afford it. So, we decided to make our own version which can be used in homes. We wanted something cheaper and easier for home use. By taking ideas from the expensive one and simplifying them, we aimed to create a friendly and affordable system for home farming.

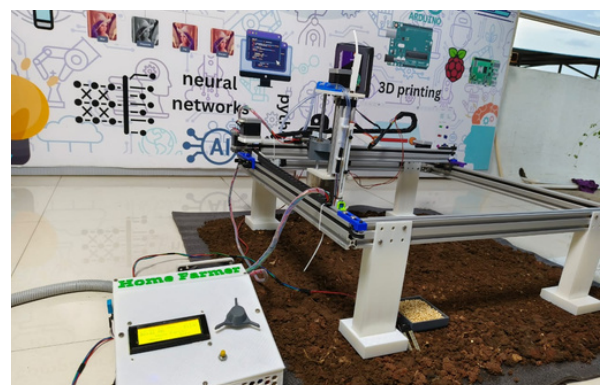
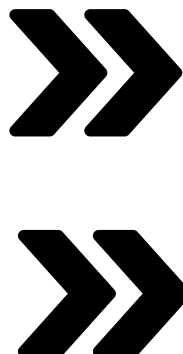
Mechanical Construction - V2



- 3D printed parts provide structural support and secure aluminum profiles.
- Aluminum profiles provide horizontal support.
- Belt and trolley provide movement for X and Y axes.
- Allen bolts connect aluminum profiles to 3D printed parts.
- Linear bearings and lead screw facilitate Z axis movement.
- Ball bearings reduce rotational friction for Z axis.
- Couplers connect stepper motors to pulley.
- The belt goes around 1 axis, with the ends connecting to the trolley, which is then moved by the stepper motors.
- Raspberry pi enables the robot to use the app and the alexa as well as the camera livestream



Picture 1.3 - Home Farmer V1



Picture 1.4 - Home Farmer V2

OUR JOURNEY - CONTINUED

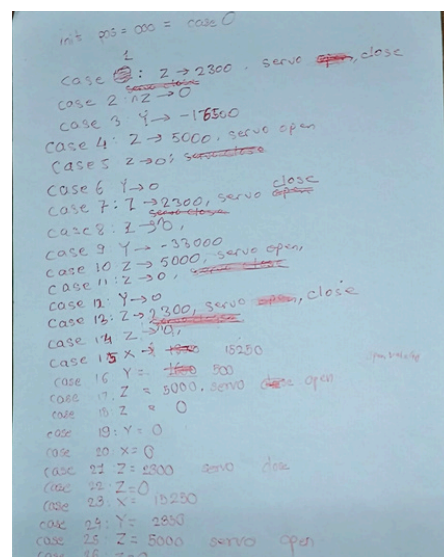


Coding of the Solution



The code automates movement with stepper motors and a servo, looping through predefined actions based on the XY-coordinate system. It's part of a robotics or automation project, requiring precise motor control. Additionally, it features a 20x4 LCD for real-time updates on the bot's function and seeding recommendations, with a switch for watering plants. This indicates its integration into a formalized project, possibly in robotics or automation, demanding meticulous motor control for synchronized operations.

```
Stepper.ino
48 void loop() {
49   // Change direction once the motor reaches target position
50
51   switch (pos) {
52     case 0:
53
54     zstepper.moveTo(-2300);
55
56     if (zstepper.distanceToGo() == 0) {
57       //yStepper.moveTo(yStepper.currentPosition());
58       pos = 0;
59       delay(2000);
60       // yStepper.moveTo(0);
61     }
62     // Move the motor one step
63     zstepper.run();
64     break;
65   case 1:
66     zstepper.moveTo(2300);
67     zstepper.run();
68
69     if (zstepper.distanceToGo() == 0) {
70       //yStepper.moveTo(yStepper.currentPosition());
71       pos = 2;
72       picker.write(40, 30, true);
73       delay(1000);
74       picker.write(0, 30, true);
75       delay(500);
76     }
77     break;
78   case 2:
79     zstepper.moveTo(0);
80
```



Scribble of
initial
code

Social Impact of the Product



The Home Farmer robot helps everyone grow fresh, healthy vegetables right at home, even in small spaces like apartments. This means people get organic, chemical-free food, reducing the need for harmful pesticides and fertilizers that damage our soil and water. By using Home Farmer, we promote healthier eating habits and protect our environment from pollution. It also encourages families and kids to learn about farming and caring for our planet. With Home Farmer, we can make our world greener and healthier, one home at a time, while making farming easy and fun for everyone!

Innovation and Challenges



Home Farmer is made special by 5 key innovations

- First – Affordability: We use 3D printing and simple components to cost much less than commercial alternatives.
- Second – Educational Value: Home Farmer turns any home into a learning lab where families can learn about sustainable farming and healthy eating firsthand.
- Third – User-Friendly: Simple design with smart sensors means minimal maintenance – meaning that even people with no gardening experience can grow fresh food!
- Fourth - Speed: We have worked on replacing the lead screw model with a faster, more efficient belt drive model.
- Fifth - Quality of Life: With its voice assistant and simple to use interface, Home Farmer helps even the people who have little to no gardening experience to grow their own food



During the construction of our robotic solution we faced many challenges. These included.....

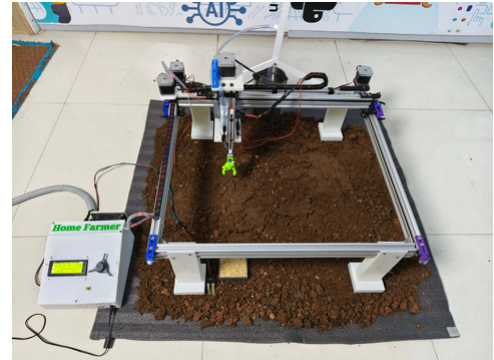
- The pump was not able to function properly and pump enough water for some time
- Problems with connecting the temperature sensor to the Raspberry Pi instead of the Arduino
- During the Alexa based planting testing we faced challenges to get it to connect to the raspberry pi
- The pulleys being too loose making the axis move get jammed.

Growing crops using Home Farmer



SCAN ME

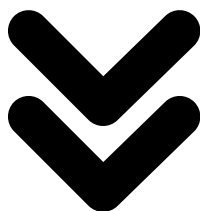
Scan the QR code to access our website, where you can find all the information, updates, as well as the coding and construction details for the Home Farmer Robot.



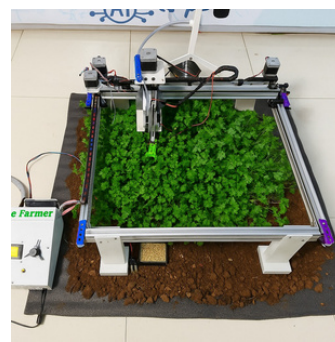
Schedule



Robot plants and waters seeds



Harvesting Time!



Entrepreneurship Aspect

Key Partners Farming and Food Related Companies	Key Activities • Designing and Prototyping • Manufacturing and assembly • User Documentation (Manuals) • Product Promotion	Key resources • Hardware • Supplier Network for Parts • Team skills (engineering, design and coding)
Value proposition • Open source for education • Irrigation using Penman's formula to avoid overwatering • Automated and easy to use	Customer relationships • Community support (forums) • Online tutorials and guides	Channels • Our own website (direct sales only) • Shipping planned through standard courier services (Note:- no distributing yet , this is a prototype stage)
Customer segments • People in urban or semi urban areas who want to grow their own fresh food as well as families	Cost structure • Hardware and electronic components • R&D and prototyping • Marketing • Customer support (Online only)	Revenue streams • Cost per unit: - INR 16000/- (Approximate) • Direct one-time unit sales

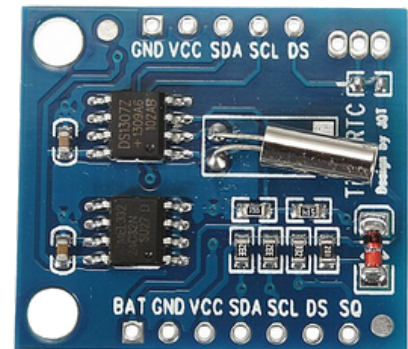
Component List

Component	Quantity	Usage of Component
DS1307 RTC	1	For Time display on LCD
20x4 I2C LCD	1	For Display of Menu, features, and for selection of action
Arduino UNO	1	Arduino UNO specifically to host the CNC Shield V3
CNC Shield V3	1	To host the Stepper Motor drivers needed
DRV 8825 Stepper Motor Driver	4	For Proper functioning of Stepper Motors
NEMA 17 Stepper Motor	4	2 for movement of x axis, 1 for y axis, and 1 for z axis.
Raspberry Pi	1	To connect the Arduino to Alexa, and the Web Dashboard.
DHT11	1	To measure the humidity and temperature of surrounding
Soil Moisture Sensor	1	To measure soil moisture for plant health
Encoder	1	For user input to select an action on LCD
Switch	1	Reset Button for emergency stop or homing
Fan (12v)	1	To cool the DRV8825 Stepper motor driver
Servo Motor (180 Degrees)	1	To pick up the seeds and plant them
Water Pump	1	To water the plants when called for

Components - Additional Information

DS1307 RTC

- I2C Address -> 0x68
- Data Pins -> SDA (A4), SCL (A5) for I2C Protocol
- Requires Li-ion coin battery for offline time backup
- Requires external library - <uRTCLib.h>



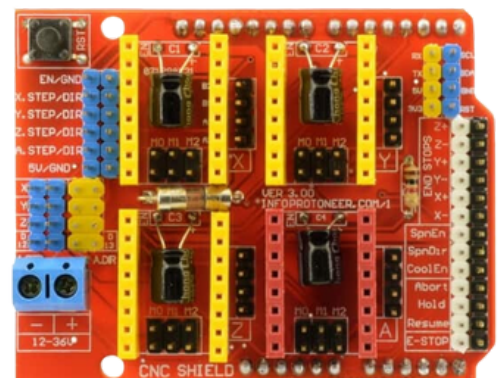
20x4 I2C LCD

- I2C Address -> 0x27
- Data Pins -> SDA (A4), SCL (A5)
- Requires external library - <LiquidCrystal_I2C.h>



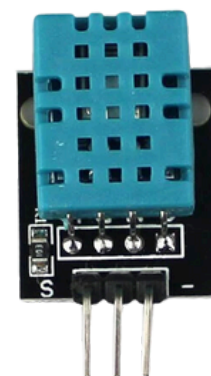
CNC Shield V3

- Supports 4 A4988 or DRV8825 Stepper motor Drivers
- Is a PCB that is placed exactly on top of the Arduino UNO board
- Supports Axis cloning



DHT11

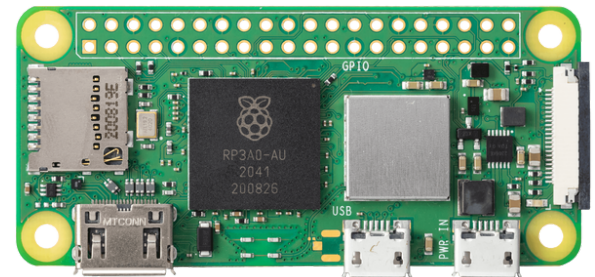
- Requires external library - <dht.h>
- Has sampling rate of 1Hz, meaning it can provide new information every 1 second.
- Temperature is given in Celsius by the DHT, but we can use the formulae to convert it into Fahrenheit or Kelvin.
- Data Pins -> Middle Pin goes to a Digital Input Pin



Components - Additional Information

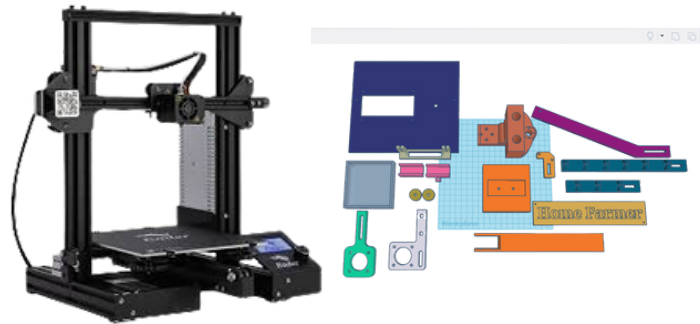
Raspberry Pi Zero 2 W

- Helps with connection to the Internet and communication with Arduino
- Camera relay to Web Dashboard



3D Printed Materials

- All Materials made on Tinkercad and Freecad and uploaded here:-
 - bit.ly/homefarmer3dprint



2040 Aluminum Profile

- Used for Basic Structure of the robot
- Is connected using T-Nut and Bolt



Lead Screw, Belt, Pulley, and Trolley

- Helps in movement of all axes
- Lead Screw allows for precision in the z axis.
- Pulley connects to stepper motor and the opposite end to host the belt, which is connected to the trolley for movement



LIST OF SOURCES

- <https://lastminuteengineers.com/ds1307-rtc-arduino-tutorial/>
- <https://lastminuteengineers.com/arduino-1602-character-lcd-tutorial/>
- https://www.youtube.com/watch?v=Hb6WJTX5X_E&pp=ygUSY25jIG1ha2luZyBtYWNoaW5l
- <https://www.youtube.com/watch?v=uNkADHZStDE&pp=ygUIZmFybSBib3Q%3D>
- <https://www.youtube.com/watch?v=tkOhH9OUMto&pp=ygUVbGVhZCBzY3JldyBhcnJhbmdlbXRu>
- <https://www.youtube.com/watch?v=zUb8tiFCwmk&pp=ygUSY25jIGFyZHVpbm8gc2hpZWxk>
- <https://www.youtube.com/watch?v=JlhjcTh4yts&pp=ygUSY25jIGFyZHVpbm8gc2hpZWxk>
- https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwjfp7noz7eFAxUas1YBHR40BXkQFnoECBEQAQ&url=https%3A%2F%2Fwww.handsontec.com%2Fdataspecs%2Fncnc-3axis-shield.pdf&usg=AOvVaw2eTWLYBw_TEDD2ZzLp-D4L&opi=89978449
- https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwjfp7noz7eFAxUas1YBHR40BXkQFnoECA4QAQ&url=https%3A%2F%2Fwww.makerstore.com.au%2Fwp-content%2Fuploads%2Ffilebase%2Fpublications%2FCNC-Shield-Guide-v1.0.pdf&usg=AOvVaw2UMJIT_IKxabxmtCtCJyOl&opi=89978449
- https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwjPzluG0LeFAxWehlYBHRRDAQ4QFnoECBQQAQ&url=https%3A%2F%2Fen.wikipedia.org%2Fwiki%2FStepper_motor&usg=AOvVaw2w8OG9jY8h_kLBLAQEAniF&opi=89978449
- <https://www.youtube.com/watch?v=eyqwLiowZiU&pp=ygUVc3RlcHBldiBtb3RvciB3b3JraW5n>
- <https://www.youtube.com/watch?v=5CmjB4WF5XA&pp=ygUac3RlcHBldiBtb3RvciBkcml2ZXIgaTQ5ODg%3D>
- <https://www.youtube.com/watch?v=BV-ouxhZaml&pp=ygUYc3RlcHBldiBtb3RvciBkcml2ZXIgaTQ5ODg%3D>
- <https://www.youtube.com/watch?v=rAIXfkRIQ5s&pp=ygUYc3RlcHBldiBtb3RvciBkcml2ZXIgaTQ5ODg%3D>
- <https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwilqtLG0LeFAXRTGcHHfSjAisQFnoECDoQAQ&url=https%3A%2F%2Flastminuteengineers.com%2Fdrv8825-stepper-motor-driver-arduino-tutorial%2F&usg=AOvVaw15iBpFG5ksEUIYtJ-z77SN&opi=89978449>
- <https://forum.arduino.cc/t/using-cnc-shield-v3-directly-with-arduino-ide/1022999/2>
- <https://www.handsontec.com/dataspecs/cnc-3axis-shield.pdf>
- <https://www.arduino.cc/reference/en/libraries/accelstepper/>
- <https://www.tinkercad.com/> - For designing parts